MIDTERM
FALL 2024

STAT 642 STATISTICAL INFERENCE
INSTRUCTOR: GARRITT PAGE

Show all work to receive credit. Credit (including partial credit) is given for work not final answers.

1. A bag contains one marble which is either green or blue, with equal probabilities. A green marble is put into the bag (so there are two marbles now), and then a random marble is taken out. The marble taken out is green. What is the probability that the remaining marble is also green?

2. Let $X \sim N(0,1)$. Find the pdf of $Y = X^4$.

3. Let $Y = X + 1$ where $X \sim Poisson(\lambda)$. Find $E(Y^2)$.

4. Is it possible to have a random variable X such that the $E(X) = 0$, $E(X^2) = 1$, $E(X^3) = 0$, $E(X^4) = 0$? Prove your claim.

5. Consider a random variable, X, with pdf

$$f(x) = \begin{cases} 4x & 0 < x < \frac{1}{2} \\ 4(1-x) & \frac{1}{2} \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find $F_X(x)$, the cdf of X, and prove that it is a valid CDF.

6. Let $\mathcal{F}$ be a σ-algebra on the sample space S. Prove that $\mathcal{F}$ is closed under countable intersection.

7. Let X have a continuous cdf $F_X(x)$ and define the random variable Y as $Y = F_X(X)$. Show that $Y \sim UN(0,1)$. That is, show that $F_Y(y) = y$.

8. Consider a random variable X with PDF,

$$f_X(x) = \begin{cases} \frac{x}{4} & 0 < x \leq 2, \\ e^{-2(x-2)} & x > 2 \\ 0 & \text{otherwise.} \end{cases}$$

Find $E(X)$ and $Var(X)$.

9. Let $X \sim Poisson(\lambda)$ and $Y \sim Poisson(\mu)$ and assume that X and Y are independent.

   (a) Show that $X + Y \sim Poisson(\lambda + \mu)$

   (b) Find the distribution of $X \mid X + Y = n$.

10. Let X and Y be *iid* $Exp(\lambda)$. Find the pdf of $U = log(X/Y)$. (Hint: parametrization of $Exp(\lambda)$ matters.)

11. If P is a probability function and A and B are any sets in some σ-algebra $\mathcal{B}$, then prove the following:

(a) $P(B \cap A^c) = P(B) - P(A \cap B)$

(b) If A and B are independent, then $P(A \cup B) = 1 - P(A^c)P(B^c)$